

BETSY GEORGE

Senior Machine Learning / AI Engineer

Dubai, UAE | +971-54-402-3702 | betsygeorge1993@gmail.com

LinkedIn: betsy-george-am2221/ | GitHub | Portfolio

PROFESSIONAL SUMMARY

Senior Machine Learning and AI Engineer with 5 years of experience designing, building, and deploying production-grade AI systems across healthcare, IoT, NLP, computer vision, and Generative AI domains. Proven track record delivering end-to-end machine learning pipelines, LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, and cloud-native inference platforms. Hands-on expertise in Python, PyTorch, TensorFlow, LangChain, FastAPI, AWS, and Docker. Experienced in fine-tuning large language models (LoRA, qLoRA), advanced prompt engineering (CoT, few-shot, in-context learning), vector database integration, and MLOps workflows including CI/CD, model monitoring, and experiment tracking. Strong collaborator across engineering, data science, product, and business stakeholder teams throughout the full software development lifecycle. Open to Senior AI Engineer, Senior ML Engineer, Generative AI Engineer, and Data Scientist roles across the UAE.

CORE COMPETENCIES

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), LangChain, LCEL, Transformer Architectures, Attention Mechanisms, LLM Fine-Tuning (LoRA, qLoRA), Prompt Engineering, Embeddings, Vector Search, Document Intelligence

Machine Learning & NLP: Predictive Modeling, Deep Learning, NLP, Text Processing, Information Retrieval, Computer Vision, Speech AI, Signal Processing, Sentiment Analysis, Classification, Clustering

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-Learn, LangChain, OpenCV, Pandas, NumPy, FAISS, Chroma, Hugging Face Transformers

Backend & APIs: Python, FastAPI, REST APIs, PostgreSQL, SQL, Apache Spark, ETL Pipelines, Data Engineering, Feature Engineering

Cloud & MLOps: AWS (EC2, Bedrock, S3), Azure, Docker, CI/CD Pipelines, Model Monitoring, Experiment Tracking, Model Versioning, ClearML, LLM Observability & Tracing

Engineering Practices: Git, Code Reviews, Automated Testing, Technical Documentation, Jupyter Notebooks, Agile Collaboration, SDLC

PROFESSIONAL EXPERIENCE

Senior Machine Learning / AI Engineer

Data Hub Technologies Pvt Ltd | Kochi, India | January 2021 – Now

- Architected and deployed production RAG systems using LangChain and LCEL — designing full pipelines for document ingestion, adaptive chunking, embedding generation, FAISS/Chroma vector indexing, and multi-stage contextual retrieval served via FastAPI with real-time inference.
- Fine-tuned large language models using LoRA and qLoRA for domain-specific use cases, optimising model behaviour and retrieval quality for production performance; established model versioning, experiment tracking, and reproducibility workflows throughout.
- Applied advanced prompt engineering techniques including chain-of-thought (CoT), few-shot, and in-context learning to improve LLM output quality; evaluated performance using Ragas-style metrics (faithfulness, answer relevancy, context precision).
- Implemented LLM observability and tracing in production environments — structured logging, monitoring dashboards, and API performance metrics — ensuring system reliability and enabling rapid debugging.
- Engineered NLP and information retrieval pipelines processing 50,000+ customer reviews, applying text processing, embedding-based retrieval, and sentiment classification to improve customer satisfaction insights by 20%.
- Productionized ML pipelines on AWS and Azure through CI/CD automation and monitoring, reducing deployment time by 25% and maintaining high availability of inference services.
- Architected IoT predictive maintenance platforms integrating sensor telemetry with ML models across edge hardware, profiling compute-constrained workloads and reducing equipment downtime by 35%.

- Developed real-time vehicle number plate recognition systems using computer vision and OpenCV pipelines, improving monitoring accuracy by 30%.
- Delivered predictive healthcare models for heart disease and diabetes achieving AUC up to 0.99 and precision of 95% using PyTorch deep learning architectures.
- Spearheaded Generative AI and speech deep learning research achieving 2.5% word error rate in voice cloning and up to 100% accuracy in speech emotion recognition.
- Built and maintained ETL data pipelines to extract, transform, and load data from diverse structured and unstructured sources into ML-ready formats for downstream model training and evaluation.
- Optimised AdaBoost ensemble inference performance, improving model efficiency by 10%; engineered unsupervised customer segmentation frameworks predicting purchasing behaviour with 75% accuracy.
- Collaborated across engineering, product, data science, and business teams throughout the SDLC — from discovery to deployment — translating complex AI capabilities into clear product features and communicating results to non-technical stakeholders.
- Led cross-functional AI workshops with 10+ stakeholders to define enterprise AI requirements, delivering a unified AI roadmap aligned with business strategy.

Research Assistant

Amrita Center for Wireless Networks | Kollam, India | August 2017 – January 2021

- Implemented ML and deep learning research algorithms in Python across NLP, signal processing, and biomedical domains; developed breast cancer detection diagnostic models achieving 96% accuracy using advanced deep learning architectures.
- Designed and ran controlled ML experiments with full tracking of hyperparameters, model versions, and evaluation metrics; improved speech recognition accuracy from 75% to 85% using DSP and deep learning techniques.
- Classified ECG signals using machine learning and signal processing pipelines, exceeding baseline performance by 7% through structured experimentation and reproducible model training workflows.
- Implemented predictive analytics-driven energy optimisation systems delivering 6% measurable energy savings.
- Produced clear technical documentation for all models, pipelines, and experiments; contributed to 2 published research outputs and won the Most Publications in Academic Year award.
- Mentored and supervised a 10-member research team, improving delivery timelines by 25% and achieving quality scores of 4.2/5.
- Collaborated with product engineering teams to streamline model validation workflows, reducing verification cycles by 50%.

KEY PROJECTS

AI Document Assistant — Production LLM + RAG System (AWS)

Enterprise-grade document intelligence platform enabling contextual Q&A from uploaded PDFs using Retrieval-Augmented Generation, deployed to production on AWS.

- Designed end-to-end RAG pipeline: document ingestion, adaptive chunking, embedding generation, FAISS/Chroma vector indexing, and multi-stage contextual retrieval with re-ranking.
- Applied LoRA-based fine-tuning for domain adaptation and advanced prompting (CoT, few-shot) to improve answer grounding and reduce hallucination rates.
- Built FastAPI backend with clean documented endpoints, automated test coverage, structured logging, and monitoring hooks for production reliability.
- Containerised application with Docker; provisioned AWS EC2 GPU infrastructure; configured reverse proxy for scalable multi-user access.

Technologies: Python | LangChain | LCEL | FAISS | Chroma | FastAPI | PostgreSQL | Docker | AWS EC2 | PyTorch | LoRA/qLoRA | Embeddings | Ragas

IoT Predictive Maintenance Platform — Edge AI & MLOps

ML-powered maintenance system integrating real-time sensor telemetry with optimised ML inference across compute-constrained IoT edge hardware.

- Architected sensor data ingestion and ETL pipelines feeding ML anomaly detection models deployed on edge accelerator hardware.

- Implemented CI/CD deployment, model versioning, and monitoring workflows; achieved 35% reduction in equipment downtime in production.

Technologies: Python | PyTorch | Scikit-Learn | AWS | Docker | CI/CD | Edge Hardware Inference | MLOps

EDUCATION

Master of Technology — Wireless Networks and Applications

Amrita School of Engineering, India | 2017

- Dissertation: Machine Learning for Intelligent Systems
- Award: Most Publications in Academic Year
- Relevant Coursework: Artificial Intelligence, Big Data Analytics, Data Science

Bachelor of Technology — Electronics and Communication Engineering

Visvesvaraya Institute of Engineering and Technology, India | 2015

CERTIFICATIONS

- Deep Learning Specialization — Coursera / DeepLearning.AI
- Generative AI Certification
- Machine Learning with Apache Spark
- IBM Data Scientist Professional Certificate
- Google Data Analytics Professional Certificate
- Applied Data Science Certification

PUBLICATIONS & RESEARCH

Published 2 peer-reviewed research papers during tenure at Amrita Center for Wireless Networks, recognised with the Most Publications in Academic Year award (M.Tech programme). Research areas: machine learning for biomedical signal processing, speech recognition, and energy optimisation systems.